

SAE FURUKAWA

[furukawa.s\[at\]northeastern.edu](mailto:furukawa.s[at]northeastern.edu)

[Google Scholar](#)

[GitHub](#)

[LinkedIn](#)

EDUCATION

Northeastern University, Boston, Massachusetts

September 2025 ~ present

Ph.D. in Computer Science; Advisor: Alina Oprea

Pomona College, Claremont, CA

May 2025

Bachelor of Arts Computer Science (GPA: 3.96); Cum Laude

RESEARCH INTERESTS

My research broadly spans privacy, security, and trustworthy AI, with recent work focusing on memorization and training-data extraction in language models.

PREPRINTS

Sae Furukawa, Alina Oprea, *Reconstruction of Personally Identifiable Information from Proprietary Data in Supervised Fine-Tuned Models*, arXiv:2605.12264, 2026. *Under submission.*

Ashley Song, **Sae Furukawa**, Zhian Zhou, Bryce Tu Chi, David Nembhard, *Machine-Learning Enhanced Human-Error Detection in Drone-Assisted Bridge Inspections*, 2025.

OTHER RESEARCH WORKS

Sae Furukawa, *Differentially Private Federated Fine-tuning of Foundation Models for Hate Speech Detection*, Pomona College Senior Thesis, advised by David Kauchak and Eleanor Birrell, 2025.

Jonat John Mathew, Rakin Ahsan, **Sae Furukawa**, Jagdish Gautham Krishna Kumar, Huzaifa Pallan, Agamjeet Singh Padda, Sara Adamski, Madhu Reddiboina, Arjun Pankajakshan, *Towards the Development of a Real-Time Deepfake Audio Detection System in Communication Platforms*, arXiv:2403.11778, 2024.

INDUSTRY EXPERIENCE

Information Security Intern | Apple Information Security (AIS), Apple, Inc

May ~ August 2024

- Developed generative models for anomaly detection in command line execution, such as Deep Autoencoding Gaussian Mixture Model (DAGMM) and Variational Autoencoder (VAE), successfully classifying all malicious commands in the top 1% of the most anomalous.
- Researched the feasibility of implementing anomaly detection through differentially private federated learning, marking AIS's first theoretical exploration of detection systems on-device.
- Employed the Moment Accountant Gaussian Mechanism to ensure privacy guarantees for federated learning models, minimizing the privacy-utility trade-off through hyperparameter tuning and maintaining model accuracy above 90%.

Software Development Intern | SMBC Nikko Securities, Tokyo, Japan

December 2023 ~ January 2024

- Developed and implemented a robust testing framework for the platform's user interface, specifically designed to support rule injections in the SNET algorithm trade system, utilizing Selenium Webdriver for automation.
- Significantly improved testing efficiency by reducing the cycle duration by 75% while expanding coverage by 400%.
- Enhanced platform UI for improved usability and user-computer interaction performance, using Groovy, CSS, and HTML.

Cybersecurity AI Enabler Intern | RediMinds, Inc

September ~ December 2023

- Built system executables aimed at recognizing and countering deep fakes and AI voice clones in real-time.
- Leveraged audio processing and feature engineering techniques, including MFCC, Mel Spectrograms, and data augmentation, to preprocess 10,000+ audio datasets and extract pertinent features.
- Designed and optimized the CNN models through hyperparameter tuning and preprocessing methods, such as normalization and learning rate scheduling, achieving a 10+% improvement in validation accuracy.
- Developed a ResNet-based model to counter voice-based logical access attacks, outperforming baseline models from the ASVspoof 2019 challenge by reducing the Equal Error Rate (EER).

TEACHING EXPERIENCE

Teaching Assistant | Department of Computer Science, Northeastern University

January 2026

- Machine Learning and Data Mining (CS4400), Prof. Alina Oprea, Spring 2026

Teaching Assistant | Department of Computer Science, Pomona College

September 2023 ~ May 2025

- Neural Networks (CS152), Prof. Anthony Clark, Spring 2025
- Algorithms (CS140), Prof. David Kauchak, Fall 2024
- Data Structures (CS62), Prof. Alexandra Papoutsaki, Fall 2023

HONORS

Grace Hopper Celebration Scholar (2023), Blackhat USA Student Scholar (2023), Math Canada National Team Candidate (2020)

SKILLS

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, HuggingFace

Programming languages: Python, C, Java/Groovy, JavaScript, CSS/HTML

Languages: English (native), Japanese (native), French (fluent)